

4.4 Measurement of β -ray spectra

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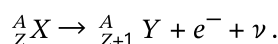
Aims and objectives

Measurements of the energies and intensities of β -rays are important in the investigation of the energy levels and characteristics (spins and parities) of the various states of atomic nuclei. In this experiment you will measure the energy spectrum of emitted β -rays from a radioactive source and determine the total energy released by one of the nuclear transitions.

Background theory

Beta-decay

Beta rays are electrons that have been ejected from the nucleus during radioactive decay. The process may be described using the following equation:



Importantly, during β decay, an electron and an anti-neutrino are produced, which share the energy released during disintegration. Consequently, for a given decay event the electron may have any kinetic energy from close to zero up to nearly the full disintegration energy, reflecting the energy sharing that occurs with the anti-neutrino. The anti-neutrino has a very low rest mass ($< 2 \times 10^{-7}$ of the electron mass) and is extremely difficult to detect experimentally (impossible with the experimental apparatus). An example momentum spectrum of the β -particles is shown in [Figure 1](#).

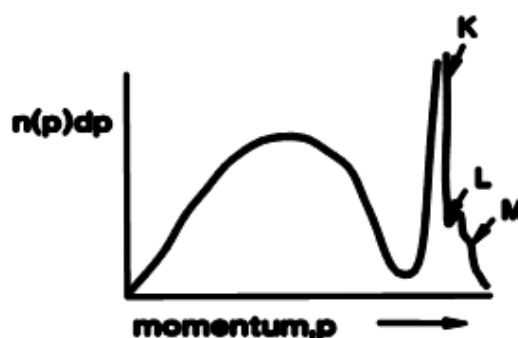


Figure 1: A representative momentum spectrum of beta particles from the decay of ^{137}Cs . The K, L and M conversion electron peaks are also depicted on the right.

In this experiment, you will be working with a source of ^{137}Cs . The decay scheme of ^{137}Cs is shown in [Figure 2](#). Here ^{137}Cs decays to an excited state of ^{137}Ba , which de-excites by either emitting either a γ -ray of energy 661.65 keV or a K conversion electron with kinetic energy 624.21 keV. The conversion line can be used for calibration of the momentum range of the spectrometer.

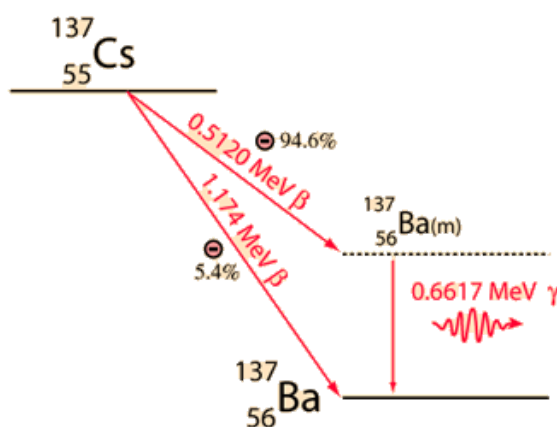


Figure 2: The decay scheme of ^{137}Cs .

Conversion electrons

Often, after radioactive decay, the nucleus of the atom can be in an excited state. One way for the atom to lose energy is by transferring the excess energy directly to an orbital electron (most often an innermost K-shell electron) where the electron is either excited to an outer shell, or is ejected from the atom and thus removed entirely. Electrons ejected via this mechanism are known as *conversion electrons* and are emitted with a characteristic well-defined kinetic energy, which makes them useful in this experiment for calibrating

the energy and momentum ranges of the spectrometer. X-rays are also produced in the process, due to electrons cascading to lower energy levels to fill the vacancy. The energy of the conversion electron emitted in this process is essentially equal to the excited state energy of the nucleus minus the binding energy of this inner shell electron to the atom.

The conversion line which you will find in the spectrum is essentially due to K electrons only. You are unlikely to resolve the L and higher conversion lines with this spectrometer. In any case the $\frac{K}{L+M}$ ratio is about 4, so little error is made by assuming that the peak is due to the K line alone.

The decay scheme of [Figure 2](#) indicates that two β -ray groups occur, both of them being twice-forbidden. The higher energy group is difficult to detect and its presence may perhaps be ignored.

The beta-ray spectrometer

Most β -spectrometers depend for their operation on the focusing properties of electric or magnetic fields. The instrument used here is a thin magnetic lens spectrometer; the effect of the magnetic field being to cause a cone of electron trajectories diverging from the source to spiral around the axis of the instrument and become a cone converging on to the detector. The focal length of such a lens depends on the electron momentum. Thus for any given magnetic field, electrons of a specific momentum are focused on the detector, whereas electrons of other momenta which would be focused at different points (compare chromatic aberration) are removed from the system by suitably placed baffles (γ -rays are also prevented from reaching the detector by a massive lead block on the axis of the instrument). Electrons of various momenta are focused by varying the magnetic field; there is an exact proportionality between the momentum of the focused electrons and the magnetic field.

The resolution of such an instrument can be made high by choosing carefully the size and shape of the baffles, only, however, at the expense of the transmission, and in practice, a compromise must be made. Typically, a thin-lens spectrometer will have a resolution of 2-3%.

The source to be investigated is of ^{137}Cs mounted as a very thin layer on a thin foil backing (to avoid undue distortion of the spectrum by back-scattering and absorption). The detector is a G-M tube with a thin mica end window.

As mentioned above, the momentum of the electrons focused on the detector is rigorously proportional to the magnetic field which, in the absence of ferromagnetic materials, is proportional to the current through the coils.

The theories of beta-decay depend on the energy of the transition, but because part of the energy of the transition has to go into creating the beta particle (contrast alpha and gamma decay), we must work with the total mass-energy, w , rather than kinetic energy, T , of the beta. Recall that:

$$w^2 = p^2c^2 + m_0^2c^4, \quad (1)$$

where p is the momentum (of the beta particle), c is the speed of light, m_0c^2 is the rest mass-energy of the electron and

$$w = T + m_0c^2. \quad (2)$$

Note that it is convenient to work in relativistic units for momentum and energy, in which momenta are in m_0c units and energies in m_0c^2 units, where m_0 is the rest mass of an electron.

Experiment

Detail of the experiment procedure are contained in the sub chapters.

Safety warning

Safety Warning:

If you are working in the laboratory near the beta ray apparatus, you are required to wear a dosimeter at all times.

You can obtain a dosimeter at the entry to the laboratory .

If you are working in person in the laboratory, radiation dosimeter readings at both the beginning and the end of the lab session are required to be reported [here](#).

Apparatus

A diagram of the β -ray spectrometer apparatus is shown below:

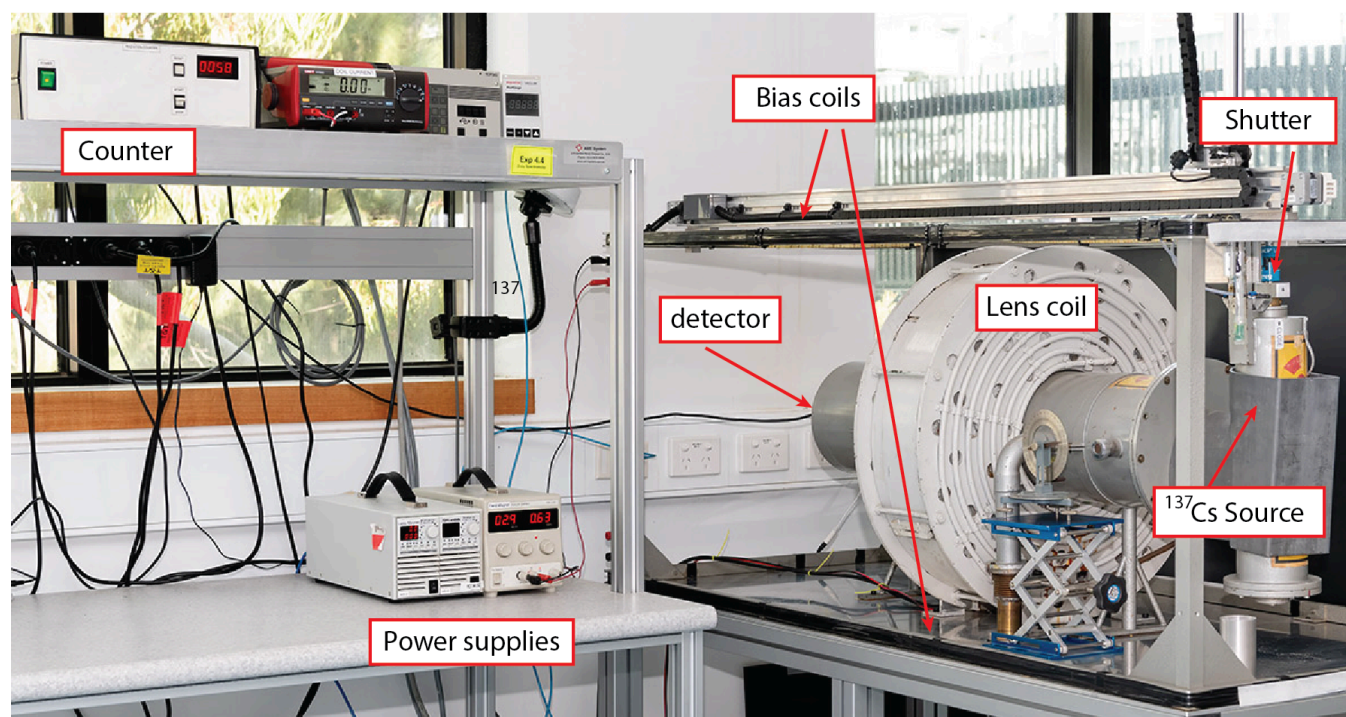


Figure 3: A labelled diagram of the beta-ray spectrometer apparatus located in the laboratory.

The β -ray spectrometer is the large tube shaped structure in the centre of the image and consists of an evacuated tube with a Cs source at one (which emits electrons by Beta decay) and detector at the end (a Geiger-Muller tube). A large externally located lens coil assembly creates a magnetic field to focus the (diverging) emitted electrons from the source onto the detector. Inside the tube a lead block stops electrons that are on axis from reaching the detector. Only electrons with a particular momentum (and hence energy) range are focused onto the detector for a given magnetic field strength (which depends on the current of the electromagnet). Hence, the momentum spectrum of the electrons (abundance of electrons as a function of momentum) can be determined by varying the current in the magnetic lens and recording the rate of detected electrons.

A detailed diagram of the β -ray spectrometer is in the [online Resources folder](#).

A pair of large, rectangular external bias coils surround the spectrometer and are driven by a controllable power supply. They are known as a Helmholtz coil pair, and are used to create a uniform magnetic field in the space between them. This coil pair can be used to cancel out the component of the Earth's magnetic field in the vertical plane. The apparatus is roughly aligned in the direction of magnetic north, and so, if the coils are used correctly, the remaining component of the Earth's field will be along the axis of the spectrometer (i.e. horizontal).

The direction of the magnetic field in the vertical plane can be visualised using the provided dip meter. When the vertical component of the Earth's field is cancelled, the dip meter pointer will be horizontal (rather than close to vertical).

The counter unit records the detection of individual electrons, and is manually controlled. Timing of the detection interval is measured by using a stopwatch.

Before you start taking data, there are some questions to think about. It is useful to remember the momentum spectrum you are expecting to see will be similar to [Figure 1](#).

1. By converting into relativistic units, where momentum $P = p/m_0c$ and energy $W = w/m_0c^2$, derive $W^2 = 1 + P^2$ from [Equation \(1\)](#).
2. What is the effect of the earth's magnetic field on the path of electrons in the spectrometer? Why is using the auxiliary Helmholtz coils, and the supplied dip meter, in your experiment useful?

3. How might you discover whether there is a constant background counting rate from the detector? Check whether there is such a background and correct for it in analysing your results.
4. You can acquire data in either constant time (time is your independent variable and the number of counts is the dependent variable) or constant count (the number of counts is the independent variable and the time is the dependent variable). You should recall from previous experiments that for a fixed time, the best estimation of the uncertainty in the number of counts, N , is $\sqrt{N}$. If you acquire to a fixed number of counts, what is the uncertainty in the measured time (due to the statistical nature of nuclear decay)?
5. Consider how you should take your data. From your knowledge of the statistics of nuclear counting (as discussed in Question 4 above), is it more sensible to count for constant times at each current setting in the spectrometer coils or to collect a roughly constant number of counts? Provide a detailed answer on the choice(s) you have decided to make and explain *why* you made them.

Hint: You may want to consider which data points of the momentum spectrum contribute the most to determining the end point energy (see the analysis section on determining the end point energy from the Kurie plot). How do you minimise the fractional uncertainty of these points? You may also want to consider how to minimise the fractional uncertainty for each data point.

Procedure

Using the bias coil power supply

To operate the power supply that provides current to the Helmholtz pair bias coils:

- Turn on the power supply marked 'field magnet'.
- Ensure that both the coarse and fine voltage control knobs are set to maximum (fully clockwise).
- Turn the 'current' knob to minimum (anti-clockwise).
- Press the 'ON' button next to the output terminals.
- Adjust the current to the desired value using the 'current' knob.

Nulling Earth's field

As per [Question 2](#), use the bias coil to remove the effect of Earth's field on the path of the electrons.

The dip meter can be used to determine that the vertical component of the Earth's field has been cancelled. It is mounted on a height and position adjustable stage. You should locate the meter at the region that corresponds to the field the electrons will take through the apparatus so that the local field is representative. The direction of the magnetic field in the plane of the meter is indicated by the double-sided pointer.

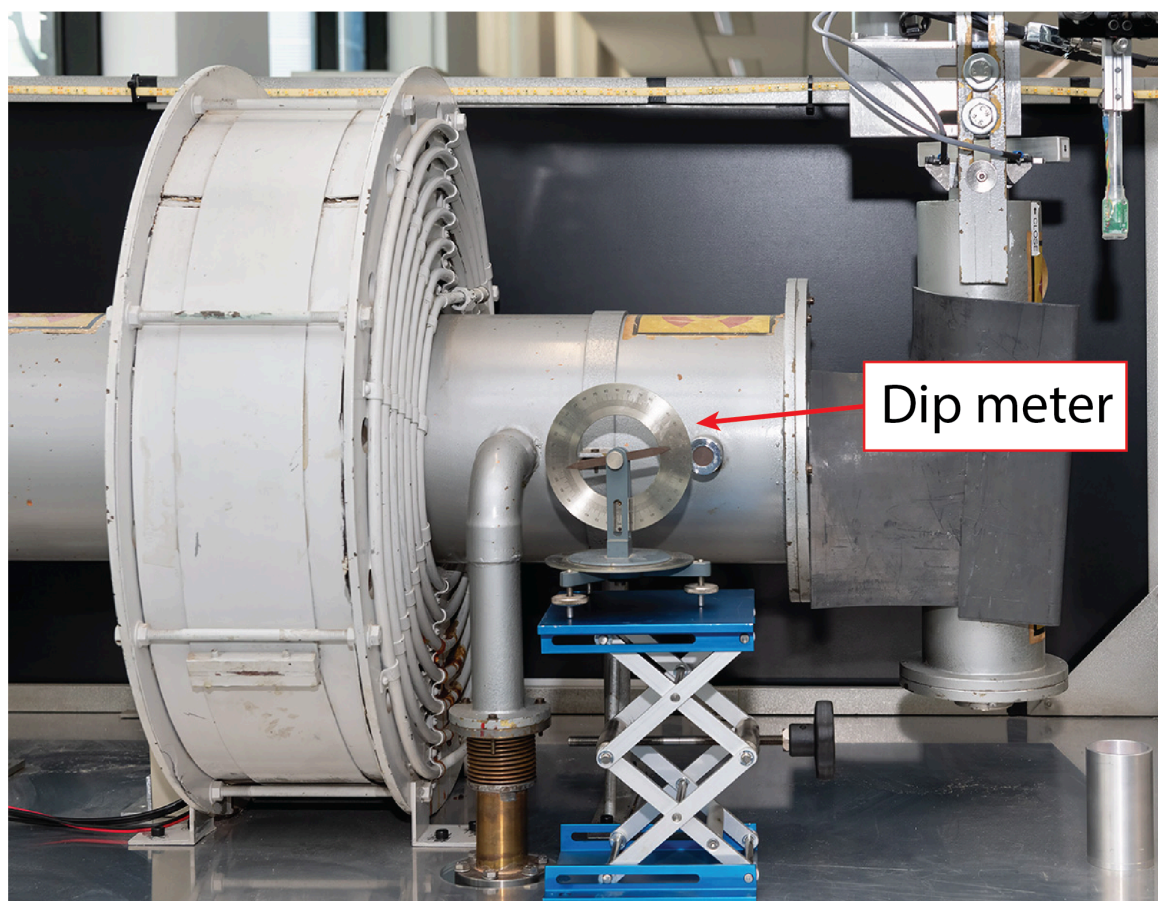


Figure 4: Side view of the apparatus showing the dip meter.

Using the lens coil power supply

This power supply provides a constant current source to the large electromagnet that forms the lens for focusing the beta rays (electrons).

To use the power supply in constant current mode:

1. Turn on power supply marked 'lens magnet'.
2. Turn the 'Current' control knob to the maximum counter-clockwise position.
3. Turn both the 'Voltage' control knobs ('Main' and 'Fine') to the maximum clockwise position.
4. Press the 'OUTPUT' button on the control panel to output current to the lens magnet. You will see that the 'cc' light is on indicating that the power supply is operating in constant current mode.
5. Adjust the 'Current' knob until the desired current is reached.

The actual current to the lens magnet should be read from the multimeter next to the counter unit marked 'coil current'.

Obtaining beta rays

The radioactive source emitting the beta rays is inside one end of the spectrometer chamber and is mounted on a rotating stage (see [Fig.3](#)). Rotating the 'shutter' lever to 'open' rotates the source to be inline with the axis of the spectrometer and allows the beta particles to enter the spectrometer and be focused.

Using the counter

The counter unit is controlled manually. To use it to count, set the toggle switch to 'run' and push the red 'reset' button. You will see the registered counts in real time on the display. To stop counting, set the toggles switch to 'stop'. Reset clears the counts.

Background radiation

As per [Question 3](#), determine whether there is a constant background counting rate so that you can account for it when analysing your results.

Acquiring the momentum spectrum and K conversion peak

As per [Question 4](#) and [Question 5](#), configure the acquisition parameters and acquire the full momentum spectrum by recording the number of counts in a time period for a variety of lens coil currents (electron momenta) as shown in [Figure 1](#).

Ensure that you plot your data as you proceed so that you can choose appropriate increments of coil current to resolve the momentum spectrum in regions where it is rapidly changing. Ensure you read the lens current using the bench multimeter located next to the counter unit.

Analysis

Information on the analysis to perform during this experiment.

Plotting the momentum spectrum

Before you plot the momentum spectrum, which is $n(p) dp$ versus p (or some equivalent form involving the coil current, which is directly proportional to p) consider the following information: the quantity $n(p) dp$ is the number of decay electrons having momentum in the range p to $p + dp$, ie having momentum in a constant range.

The spectrometer's resolution is a function of momentum; in fact it has a constant fractional resolution, i.e., $\frac{\Delta p}{p}$ is constant. Thus the momentum range Δp accepted and counted at low coil currents is smaller than that at higher coil currents. This means that a plot of raw counting rate as a function of coil current is not a true momentum spectrum. How may the true momentum spectrum be derived from your data? Consider consulting [Siegbahn, page 82,83](#) (available in the online Resources folder).

Hint: One way to visualise the situation is to think of your spectrum as a histogram, showing the number of counts in each momentum bin. For a histogram, each bin corresponds to a particular range of values that are accepted, and the widths of each bin are all the same (this would correspond to counts due to fixed range of momenta in our experiment). Here, because of how the spectrometer operates, the widths of the bins becomes wider as the momentum values increase. To convert your histogram to a spectrum you need to remove the effect of the bin widths increasing as the momentum increases.

The Kurie plot

This is the standard method for determining end-points of β -ray groups and for examining the degree of forbidden-ness of the transitions. You will be using a Kurie plot to investigate the spectrum you have measured. The momentum spectrum may be written as:

$$n(p)dp = K_1 F(Z, w) p^2 (w_0 - w)^2 S_n(w) dp, \quad (3)$$

where p is the electron momentum, w is the total energy of the electron and w_0 the total energy released in the nuclear transition. $F(Z, w)$ is the Coulomb field factor or Fermi function and a related function (or a modified function) is tabulated as a function of p (see below). $S_n(w)$ is the so-called shape factor, whose form depends on the degree of forbidden-ness of the transition. For an allowed transition $S_0(w) = 1$.

Modified Fermi function

A function related to $F(Z, w)$ is tabulated and can be found in the [MonashSPA Python library](#) (it is also available in the [resources folder](#) on Moodle [\[direct download\]](#) if you wish to perform your analysis in different software). This modified function is

$G = \frac{pF(Z=55, w)}{w}$. To import the data into your Python script:

```
import monashspa.PHS3000 as spa
fermi_data = spa.betaray.modified_fermi_function_data
```

Here the data imported from MonashSPA is a 2D numpy array, where the first column are values of p in relativistic units of m_0c^2 and the second column is the value of G for the corresponding p .

You will need to interpolate this data in order to find G for the each of the p values you have. To interpolate in Python, add to your script:

```
from scipy.interpolate import interp1d
interpolated_fermi_data = interp1d(fermi_data[:,0], fermi_data[:,1], kind='cubic')
```

You can then evaluate the interpolated fermi data for a set of values of p . For example, if you have a numpy array p containing your set of p values from your experiment, you can then do:

```
my_G = interpolated_fermi_data(p)
```

to produce a set of corresponding G values. Full documentation for interpolation is available on the SCIPY website.

Linearise your data

We can then rewrite Equation (3) in terms of G so that it becomes:

$$\left(\frac{n}{pwG}\right)^{\frac{1}{2}} = K_2 (S_n(w))^{\frac{1}{2}} (w_0 - w) \quad (4)$$

It is clear that for an allowed transition, a plot of $\left(\frac{n}{pwG}\right)^{\frac{1}{2}}$ against w yields a straight line intersecting the w -axis at w_0 . Forbidden transitions may lead to a non linear Kurie plot, which still yields the correct end point. Providing the non-linearity is not too marked, the end point may still be determined accurately. Determine the end point of your spectrum in this way.

The Shape factor

Investigate the results of using the shape factor given in [Siegbahn, page 497](#).

From your determined end point energy, calculate the corresponding beta particle kinetic energy, and compare to the accepted value of the (maximum) beta particle energy for ^{137}Cs .

References

1. [K. Siegbahn \(ed\), " \$\alpha\$ -, \$\beta\$ - and \$\gamma\$ -ray Spectroscopy", Vol 1. In particular, pages 82,83, 145-150, 495- 498.](#)

Other references not mentioned in the text

W.E. Meyerhof, "Elements of Nuclear Physics", Section 4-6.

H.A. Enge, "Introduction to Nuclear Physics", chapter 11.